Physics 5313: Electromagnetic Theory 2 (Fall 2026)

Lecture 17: Optical Fibers 2

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Transmission of meridional rays or equivalently to rays in a "'slab" geometry

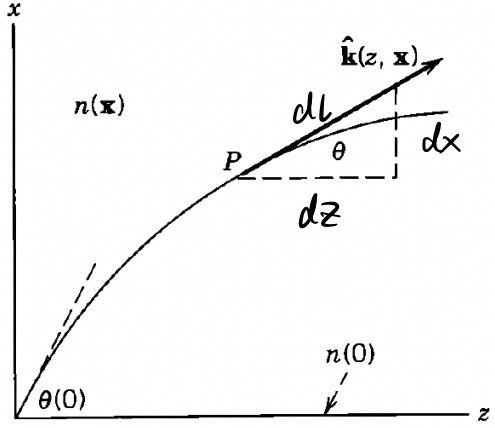
Generalized Snell's Eq.

$$\frac{d}{dl}\left[n(\vec{x}) \frac{d\vec{x}}{dl} \right] = \nabla n(\vec{x})$$

$$\frac{d\vec{x}}{dl} = (\sin\theta, 0, \cos\theta)$$

$$\nabla n = \left(\frac{dn}{dx}, 0, 0 \right)$$

$$\Rightarrow \begin{cases} \frac{d}{dl}[n\sin\theta] = \frac{dn}{dx} & ① \\ n\cos\theta = \bar{n} & ② \end{cases}$$



Ray tracing

$$\cos\theta = \frac{dz}{dl} \qquad \sin\theta = \frac{dx}{dl}$$

② $\Rightarrow n \frac{dz}{dl} = \bar{n}$

$$\frac{d}{dl} = \frac{dz}{dl}\frac{d}{dz} = \frac{\bar{n}}{n}\frac{d}{dz} \quad ③$$

③ $\Rightarrow$ ①

$$\frac{\bar{n}}{n}\frac{d}{dz}\left(n\frac{\bar{n}}{n}\frac{dx}{dz}\right) = \frac{dn}{dx}$$

$$\Rightarrow \quad \frac{\bar{n}^2}{n}\frac{d^2x}{dz^2} = \frac{dn}{dx}$$

Ray tracing

$$\Rightarrow \quad \bar{n}^2 \frac{d^2x}{dz^2} = n\frac{dn}{dx} = \frac{1}{2}\frac{d(n^2)}{dx}$$

$$\Rightarrow \quad \bar{[} \qquad , \qquad \bar{]} \frac{dx}{dz}$$

$$\Rightarrow \quad \bar{n}^2 \frac{dx}{dz}\frac{d^2x}{dz^2} = \frac{1}{2}\frac{d(n^2)}{dz}$$

$$\bar{n}^2 \cdot \frac{1}{2}\frac{d}{dz}\bar{[}\left(\frac{dx}{dz}\right)^2\bar{]} = \frac{1}{2}\frac{d(n^2)}{dz}$$

$$\bar{n}^2 \frac{d}{dz}\bar{[}\left(\frac{dx}{dz}\right)^2\bar{]} = \frac{d(n^2)}{dz}$$

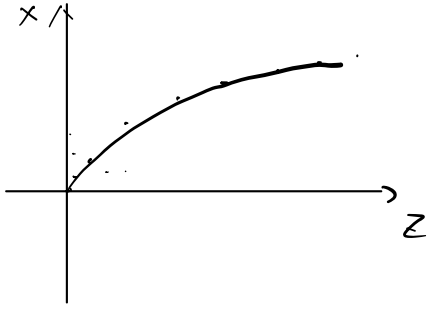
Ray tracing

$$\int \left[\bar{n}^{-2} \frac{d}{dz} \left[\left(\frac{dx}{dz} \right)^2 \right] = \frac{d(n^2)}{dz} \right] dz$$

$$\bar{n}^2 \left(\frac{dx}{dz} \right)^2 = n^2 - \bar{n}^2$$

$$\frac{dx}{dz} = \frac{(n^2 - \bar{n}^2)^{1/2}}{\bar{n}}$$

$$\frac{dz}{dx} = \frac{\bar{n}}{(n^2 - \bar{n}^2)^{1/2}}$$



$$z = \bar{n} \int_0^x \frac{1}{(n^2 - \bar{n}^2)^{1/2}} dx = z(x)$$

Ray tracing

Half period of the ray

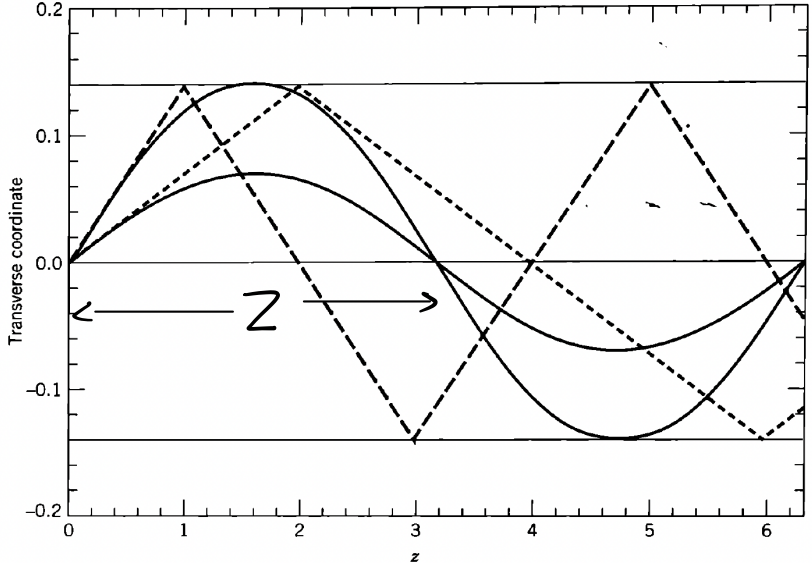
$$Z = 2\bar{n} \int_0^{x_{max}} \frac{dx}{(n^2 - \bar{n}^2)^{1/2}}$$

dashed:

Step index

Solid:

graded



Ray tracing

$$L_{phy} = \int_A^B dL \qquad L_{opt} = \int_A^B n\, dL$$

Physical path length

Optical path length

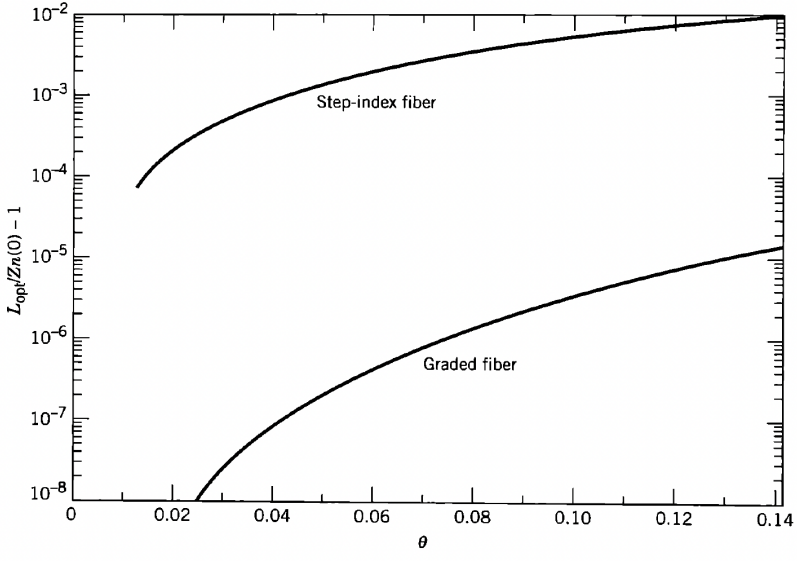
$$\frac{dz}{dx} = \frac{\bar{n}}{(n^2 - \bar{n}^2)^{1/2}} \Rightarrow dL = \sqrt{dx^2 + dz^2} = \frac{n}{(n^2 - \bar{n}^2)^{1/2}}\, dx$$

Transit time : $\bar{T} = \frac{L_{opt}}{Z}\frac{s}{c}$

s is the total length of the optical fiber.

Ray tracing

$n(x)$? Design $n(x)$ so that $\frac{L_{opt}}{Z}$ is a constant.



What is eikonal?

$$S \; ? \qquad \nabla S = n \hat{k}$$

$$\Rightarrow \quad \frac{dS}{d\vec{x}} = n \frac{d\vec{x}}{dL}$$

$$\Rightarrow \quad dS = n \frac{d\vec{x}}{dL} \cdot d\vec{x} = n \, dL$$

$$S = \int_A^B n \, dL = L_{opt}$$

S is optical path length !

Optical fibers

- Two classes:
  - Multimode
    - Cores are typically 50 μm in diameter.
    - The diameter is much larger than the wavelength of light (<1 μm)
    - Semi-geometrical optics.
  - **Single mode**
    - Cores are typically 5 μm in diameter, which is comparable with wavelength of light.

Wave Equations

$$\nabla^2\mathbf{E} + \mu_0\omega^2\epsilon(\mathbf{x})\mathbf{E} + \nabla\left(\frac{1}{\epsilon}\mathbf{E}\cdot\nabla\epsilon\right) = 0$$
$$\nabla^2\mathbf{H} + \mu_0\omega^2\epsilon(\mathbf{x})\mathbf{H} - i\omega(\nabla\epsilon)\times\mathbf{E} = 0 \qquad (8.108)$$

$$\frac{\epsilon}{\epsilon_0} = n^2$$

$$\Rightarrow \begin{cases} \nabla^2\vec{E} + \frac{n^2\omega^2}{c^2}\vec{E} = -\nabla\left(\frac{1}{n^2}(\nabla n^2)\cdot\vec{E}\right) \\ \nabla^2\vec{H} + \frac{n^2\omega^2}{c^2}\vec{H} = i\omega\epsilon_0(\nabla n^2)\times\vec{E} \end{cases}$$

Example: Equations for Hz and Ez

$\vec{E}, \vec{H} \propto e^{ikz - i\omega t}$

E_z component:

$$\nabla^2 E_z + \frac{n^2\omega^2}{c^2} E_z = -\nabla\left(\frac{1}{n^2}(\nabla n^2)\cdot\vec{E}\right)\cdot\hat{z}$$

$$LHS = \nabla_t^2 E_z + \frac{\partial^2}{\partial z^2} E_z + \frac{n^2\omega^2}{c^2} E_z$$

$$= \nabla_t^2 E_z - k^2 E_z + \frac{n^2\omega^2}{c^2} E_z \quad \text{Let } \gamma^2 = \frac{n^2\omega^2}{c^2} - k^2$$

$$= \nabla_t^2 E_z + \gamma^2 E_z$$

Example: Equations for Hz and Ez

$$RHS = -\nabla\left(\frac{1}{n^2}(\nabla n^2)\cdot\vec{E}\right)\cdot\hat{z}$$

Let's assume $\frac{\partial n^2}{\partial z} = 0 \Rightarrow \nabla n^2 = \nabla_t n^2$

$$RHS = -\frac{\partial}{\partial z}\left(\frac{1}{n^2}\nabla_t n^2 \cdot \vec{E}\right)$$

$$= -\frac{1}{n^2}\nabla_t n^2 \cdot \frac{\partial \vec{E}}{\partial z} = -\frac{ik}{n^2}\nabla_t n^2 \cdot \vec{E}$$

$$= -\frac{ik}{n^2}\nabla_t n^2 \cdot \vec{E_t}$$

Example: Equations for Hz and Ez

In WG, based on the Maxwell Eq:

$$\vec{E_t} = \frac{i}{\gamma^2}\left[k\nabla_t \bar{E}_z - \omega\mu_0 \hat{z} \times \nabla_t \bar{H}_z \right]$$

$$RHS = \frac{k}{\gamma^2}\left[k\nabla_t \bar{E}_z - \omega\mu_0 \hat{z} \times \nabla_t \bar{H}_z \right] \cdot \frac{\nabla_t n^2}{n^2}$$

$$= \frac{k^2}{\gamma^2 n^2} \nabla_t n^2 \cdot \nabla_t \bar{E}_z - \frac{\omega\mu_0 k}{\gamma^2 n^2} \nabla_t n^2 \cdot (\hat{z} \times \nabla_t \bar{H}_z)$$

$$= \frac{k^2}{\gamma^2 n^2} \nabla_t n^2 \cdot \nabla_t \bar{E}_z + \frac{\omega\mu_0 k}{\gamma^2 n^2} \hat{z} \cdot (\nabla_t n^2 \times \nabla_t \bar{H}_z)$$

Equations for Hz and Ez

Assume $n(\rho)$

$$\nabla_t^2 H_z + \gamma^2 H_z - \left(\frac{\omega}{\gamma c}\right)^2 (\nabla_t n^2) \cdot \nabla_t H_z = -\frac{\omega k_z \epsilon_0}{\gamma^2} \hat{\mathbf{z}} \cdot [\nabla_t n^2 \times \nabla_t E_z]$$

$\hat{\rho}$ $\hat{\rho}$

and

$$\nabla_t^2 E_z + \gamma^2 E_z - \left(\frac{k_z}{\gamma n}\right)^2 (\nabla_t n^2) \cdot \nabla_t E_z = \frac{\omega k_z \mu_0}{\gamma^2 n^2} \hat{\mathbf{z}} \cdot [\nabla_t n^2 \times \nabla_t H_z]$$

- Not Laplace Equations anymore. Ez and Hz are coupled together unless there is no azimuthal variation.
- Hybrid mode: Ez!=0 and Hz!=0.

$\psi(\rho)$

Example: Modes in Step-Index Circular Fibers

$$n = \begin{cases} n_1 & \rho < a \\ n_2 & \rho > a \end{cases}$$

$$\nabla_t n^2 = -2 n_1^2 \Delta \, \delta(\rho - a) \, \hat{\rho}$$

Example: Modes in Step-Index Circular Fibers

$$\begin{Bmatrix} E_z \\ H_z \end{Bmatrix} = \begin{Bmatrix} A_e \\ A_h \end{Bmatrix} J_m(\gamma\rho) e^{im\phi}, \qquad \rho < a$$

$$\begin{Bmatrix} E_z \\ H_z \end{Bmatrix} = \begin{Bmatrix} B_e \\ B_h \end{Bmatrix} K_m(\beta\rho) e^{im\phi}, \qquad \rho > a \qquad \beta^2 = k^2 - \frac{n_2^2 w^2}{c^2}$$

$\rho = a$ BC: $$\left(\frac{n_1^2}{\gamma}\frac{J_m'}{J_m} + \frac{n_2^2}{\beta}\frac{K_m'}{K_m}\right)\left(\frac{1}{\gamma}\frac{J_m'}{J_m} + \frac{1}{\beta}\frac{K_m'}{K_m}\right) = \frac{m^2}{a^2}\left(\frac{n_1^2}{\gamma^2} + \frac{n_2^2}{\beta^2}\right)\left(\frac{1}{\gamma^2} + \frac{1}{\beta^2}\right)$$

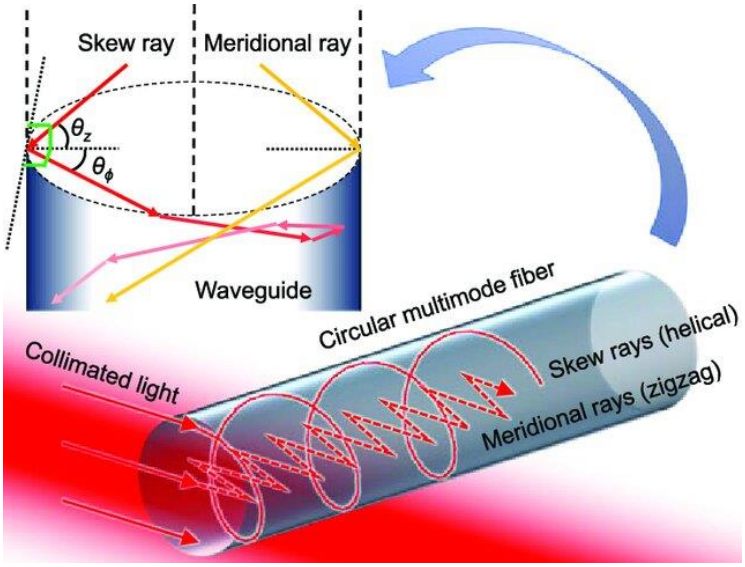
Example: Modes in Step-Index Circular Fibers

TE and TM mode (m=0):

- Lowest cutoff frequency:
- V=$n_1 \omega a\sqrt{2\Delta}/c$=2.405
- $\omega = 2.405c/(n_1 a\sqrt{2\Delta})$
- Meridional rays

Hybrid mode (EH): m>0

- No “cutoff” frequency.
- All frequencies can propagate.
- Skew ray modes



Example: Modes in Step-Index Circular Fibers

- Meridional rays: rays that pass through the cylinder axis; they correspond to modes with vanishing azimuthal index m and nonvanishing intensity at r=0
- Skew rays: rays that originate off-axis and whose path is a spiral in space with inner and outer turning points in radius; they correspond to modes with nonvanishing azimuthal index m and vanishing intensity at r = 0.